



Multi-Modal Classification of Images and Text

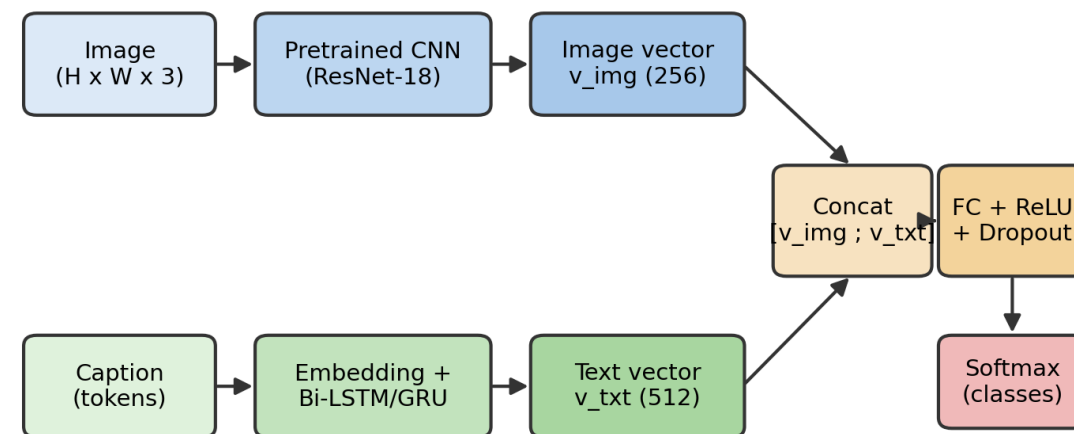
Fusing a CNN and an RNN for fashion-item classification

Deep Learning Project • 2024–2025

Member 1 • **Member 2** • **Member 3**

■ Problem & Objectives

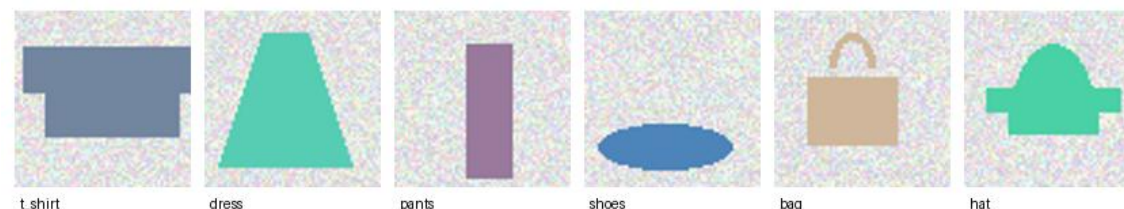
- Fashion items come with BOTH a photo and a text description.
- Using only one modality throws away useful information.
 - A blurry photo may still have a clear caption — and vice-versa.
- Goal: one model that uses image + text to classify the item.
- Objectives from the brief:
 - Classify images with a CNN
 - Encode captions with embeddings + an RNN
 - Fuse both modalities for the best accuracy
 - Describe the optimal model mathematically



Our multi-modal pipeline at a glance

■ Dataset: Fashion-Gen

- Fashion-Gen: clothing photos paired with style/category text.
- The image gives the visual signal; the caption gives the textual signal.
- Category label (e.g. SHOES, DRESSES) is the classification target.
- Reproducible demo set with the SAME structure ships with the code:
 - 6 categories, learnable shapes + templated captions
 - Lets us run the full pipeline in ~2 min, no big download
 - One modality is corrupted per sample → lets us measure fusion



Sample images from the demo set (one per class)

■ Model Architecture

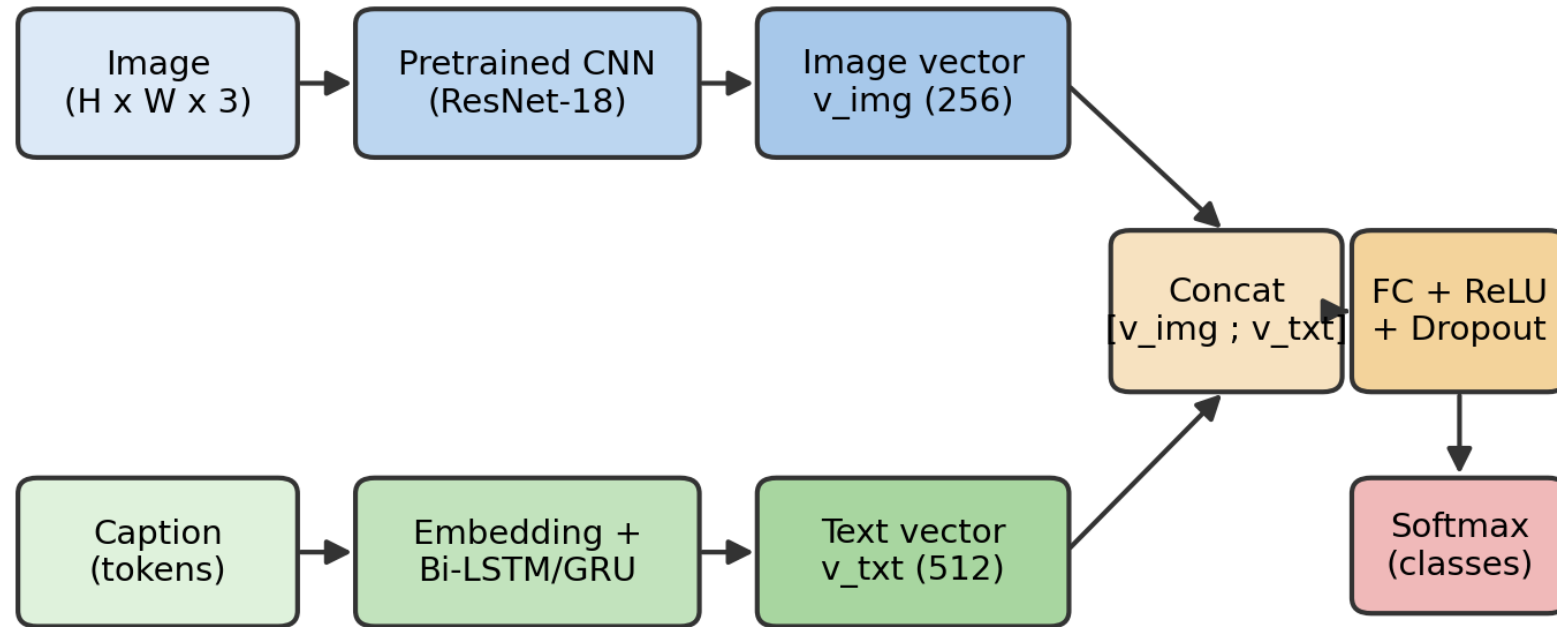


Figure 1 — image branch (CNN) + text branch (RNN) + concatenation fusion + softmax

- Two encoders produce one vector each → concatenated → FC layers → softmax over classes.

■ Image Branch — CNN

- Convolution = learnable filters slide over the image, detecting local patterns.
 - Weight sharing → few parameters, translation-equivariant
- ReLU adds the non-linearity; max-pooling down-samples.
- Transfer learning: ResNet pre-trained on ImageNet.
 - Drop its 1000-class head, keep the backbone as a feature extractor
 - Residual (skip) connections let very deep nets train
- Output: a 256-dim image vector v_{img} .

$$S(i, j) = (I * K)(i, j) = \sum_{m=0}^{k-1} \sum_{n=0}^{k-1} I(i+m, j+n) K(m, n) + b$$

$$\text{ReLU}(x) = \max(0, x)$$

$$y_{i,j} = \max_{(p,q) \in R_{i,j}} x_{p,q}$$

$$y = \mathcal{F}(x, \{W_i\}) + x$$

■ Text Branch — Embeddings + RNN

- Each word $\rightarrow$ dense embedding vector (trainable, or GloVe).
- An RNN reads the words and keeps a hidden state.
- We use an LSTM: a memory cell with 3 gates.
 - forget = what to keep, input = what to add, output = what to read
 - Additive cell update $\rightarrow$ no vanishing gradient $\rightarrow$ long context
- Bidirectional $\rightarrow$ uses past AND future words.
- GRU is a lighter 2-gate alternative.

$$e_t = E^\top x_t \in \mathbb{R}^d$$

$$f_t = \sigma(W_f[h_{t-1}, e_t] + b_f) \quad (\text{forget gate})$$

$$i_t = \sigma(W_i[h_{t-1}, e_t] + b_i) \quad (\text{input gate})$$

$$o_t = \sigma(W_o[h_{t-1}, e_t] + b_o) \quad (\text{output gate})$$

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, e_t] + b_c) \quad (\text{candidate})$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (\text{new cell state})$$

$$h_t = o_t \odot \tanh(c_t) \quad (\text{new hidden state})$$

■ Multi-Modal Fusion

- Concatenate the two vectors into one joint vector z .
- Pass z through fully-connected (dense) layers + ReLU + dropout.
- Each FC unit sees features from BOTH modalities at once.
 - $\rightarrow$ it can learn cross-modal correlations
 - e.g. fire when image looks like a shoe AND caption says 'shoes'
- This is what lets fusion beat either single modality.

$$z = [v_{img}; v_{txt}] \in \mathbb{R}^{d_{img} + d_{txt}}$$

$$h = \text{Dropout}(\text{ReLU}(W_1 z + b_1))$$

$$p_k = \frac{e^{z_k}}{\sum_{j=1}^C e^{z_j}}, \quad k = 1, \dots, C$$

■ Loss & Optimisation

- Softmax turns logits into class probabilities.
- Cross-entropy loss = $-\log(\text{prob of the correct class})$.
 - Small when confidently correct, large when wrong
- We minimise it with Adam (adaptive gradient descent).
 - Keeps running mean & variance of gradients
 - Per-parameter step size $\rightarrow$ fast, robust
- Learning-rate step decay for fine-tuning near the end.

$$\mathcal{L} = - \sum_{k=1}^C y_k \log p_k = -\log p_y$$

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

$$\theta_t = \theta_{t-1} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

■ Regularisation & Validation

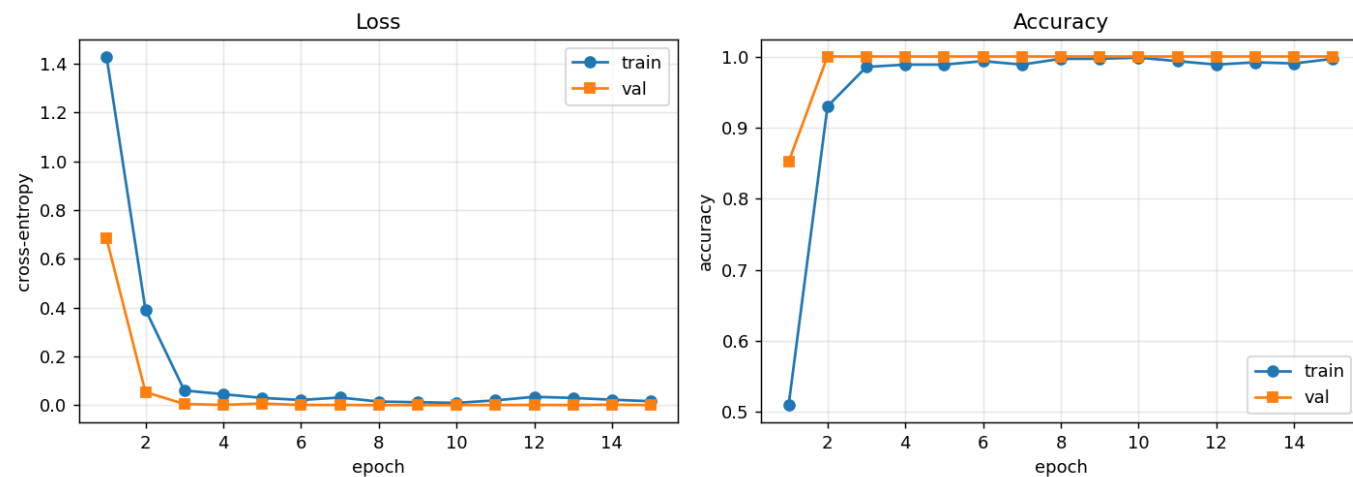
- Dropout in the fusion layers — randomly zero units while training.
 - Stops the model relying on any single feature
- Weight decay (L2) + light image augmentation.
- k-fold cross-validation to check generalisation.
 - Our 5-fold CV: mean 100.0%, std 0.0 pts → very stable
- Random search to tune LR, batch size, RNN type, dropout, FC layers.

$$\tilde{h} = \frac{1}{q} (r \odot h), \quad r_i \sim \text{Bernoulli}(q)$$

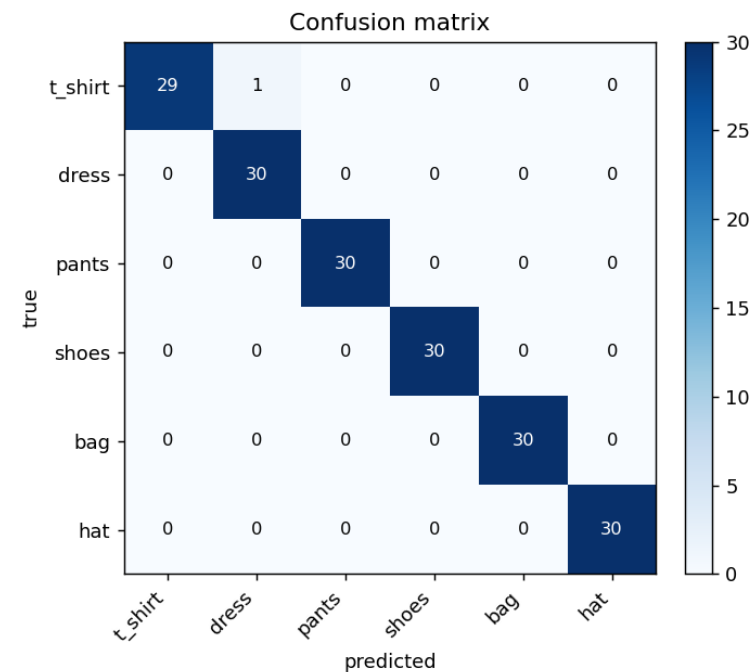
100.0%

5-fold cross-validation accuracy

Results — Training & Confusion



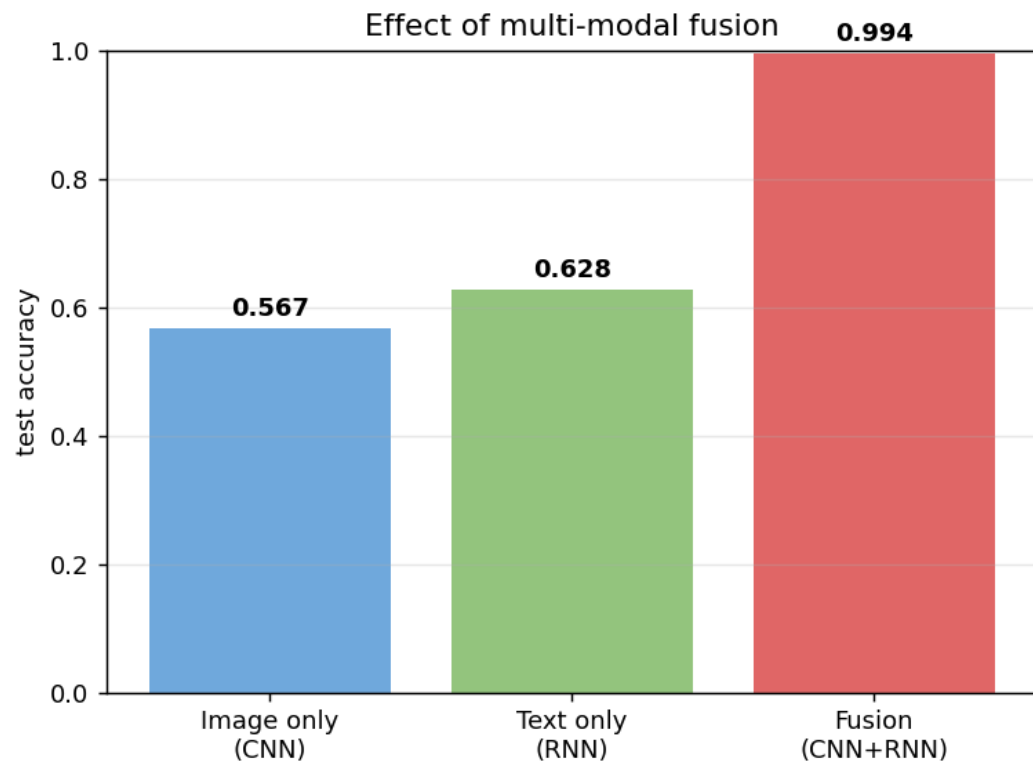
Loss & accuracy per epoch (train vs val)



Confusion matrix (test set)

Fused model: best val 100.0% | test accuracy 99.4%

■ Does Fusion Help? (Ablation)



56.7% Image only (CNN)

62.8% Text only (RNN)

99.4% Fusion (CNN + RNN)

Each modality alone is limited (one modality is corrupted per sample).
The FC fusion layer recovers the information neither carries alone.

■ Conclusion

- We built and mathematically described a CNN + RNN multi-modal classifier.
- Transfer learning (ResNet) + a gated RNN (LSTM/GRU) + concatenation fusion.
- Fusion lifted test accuracy from 56.7% (image) and 62.8% (text) to 99.4% (fused).
- The fully-connected fusion layer is what captures the image–text correlations.
- Future work: attention-based fusion, full Fashion-Gen training.

Thank you — Questions?